

Durability of Glycemic Response with FDC of Sitagliptin and Metformin in Treatment-naïve T2DM Indian Patients from a Real-world Retrospective EMR-based Study

Pranav Sheth^{*1}, Parag Shah², Prachi Ahire³, Bharadwaja Pendurthi¹, Shruti Dharmadhikari¹, Chintan Khandhedia¹, Neeraj Markandeywar¹, Amey Mane¹, Suyog Mehta¹
¹Sun Pharma Laboratories Ltd., Mumbai, India; ²Gujarat Endocrine Centre, Ahmedabad, India; ³Sun Pharmaceutical Industries Ltd., Mumbai, India; ^{*}Presenting author

INTRODUCTION

- ▶ Type 2 diabetes mellitus (T2DM) poses a serious public health challenge due to its global burden. In 2024, 588.7 million people worldwide are estimated to have diabetes¹
- ▶ The prevalence of diabetes in India is also considerably higher, currently estimated to be 101.3 million T2DM patients²
- ▶ Metformin is generally the first line therapy in T2DM but may be inadequate in patients with high HbA1c levels to achieve glycemic control^{3,4}
- ▶ Sitagliptin, a dipeptidyl peptidase 4 (DPP-4) inhibitor, has shown significant glycemic reduction. It has beta cell preserving action and is weight neutral. It is a useful treatment option for patients with T2DM⁵
- ▶ Fixed drug combinations (FDCs), such as that of metformin and sitagliptin, offer several advantages, including improved treatment adherence, simplified dosing, and potentially enhanced efficacy⁶
- ▶ American Diabetes Association (ADA) Standards of Care in Diabetes 2025 guidelines recommend that combination therapy can be considered in adults with T2DM at treatment initiation to shorten attainment of individualized glycemic goals⁷
- ▶ Clinical trials have demonstrated the efficacy of this FDC⁶, however, real-world evidence on its durability and effectiveness in treatment-naïve Indian patients is limited

AIM / OBJECTIVE

- ▶ This real-world, retrospective observational study using electronic medical records (EMR) data was conducted to evaluate the efficacy of sitagliptin and sitagliptin + metformin FDC in Indian patients with T2DM
- ▶ This subgroup analysis assessed the durability of glycemic control with sitagliptin + metformin FDC among treatment naïve patients from baseline to 3 months and 9 months

MATERIALS AND METHODS

Study Design

- ▶ In this study, aggregated and anonymized EMR data from January 2017 to March 2023 of the patients meeting the eligibility criteria was retrieved and analyzed
- ▶ Study was approved by Independent Ethics Committee (approval number: RPIEC210923) and registered on Clinical Trial Registry of India (CTRI/2023/10/058366)

Inclusion Criteria

- ▶ Male and female patients with T2DM ≥18 years of age
- ▶ Patients prescribed with sitagliptin alone or with other oral antidiabetic medications, or patients prescribed with sitagliptin + metformin FDC alone or with other oral anti-diabetic medications
- ▶ Patients with EMR data captured for more than one visit

Exclusion Criteria

- ▶ Patients with type-1 diabetes mellitus
- ▶ Patients on insulin or any other injectable antidiabetic medication like GLP -1 agonists
- ▶ Patients with incomplete medical records with respect to the required study parameters

Primary Endpoint

- ▶ Mean change in HbA1c in patients with baseline HbA1c ≥7% in sitagliptin + metformin FDC arm from baseline to 3 months

Key Secondary Endpoints

- ▶ Mean change in HbA1c in patients with baseline HbA1c ≥7% in sitagliptin + metformin FDC arm from baseline to 9 months
- ▶ Proportion of patients achieving HbA1c<7% at 3 months
- ▶ Proportion of patients achieving HbA1c<7% at 9 months

RESULTS

Follow-up

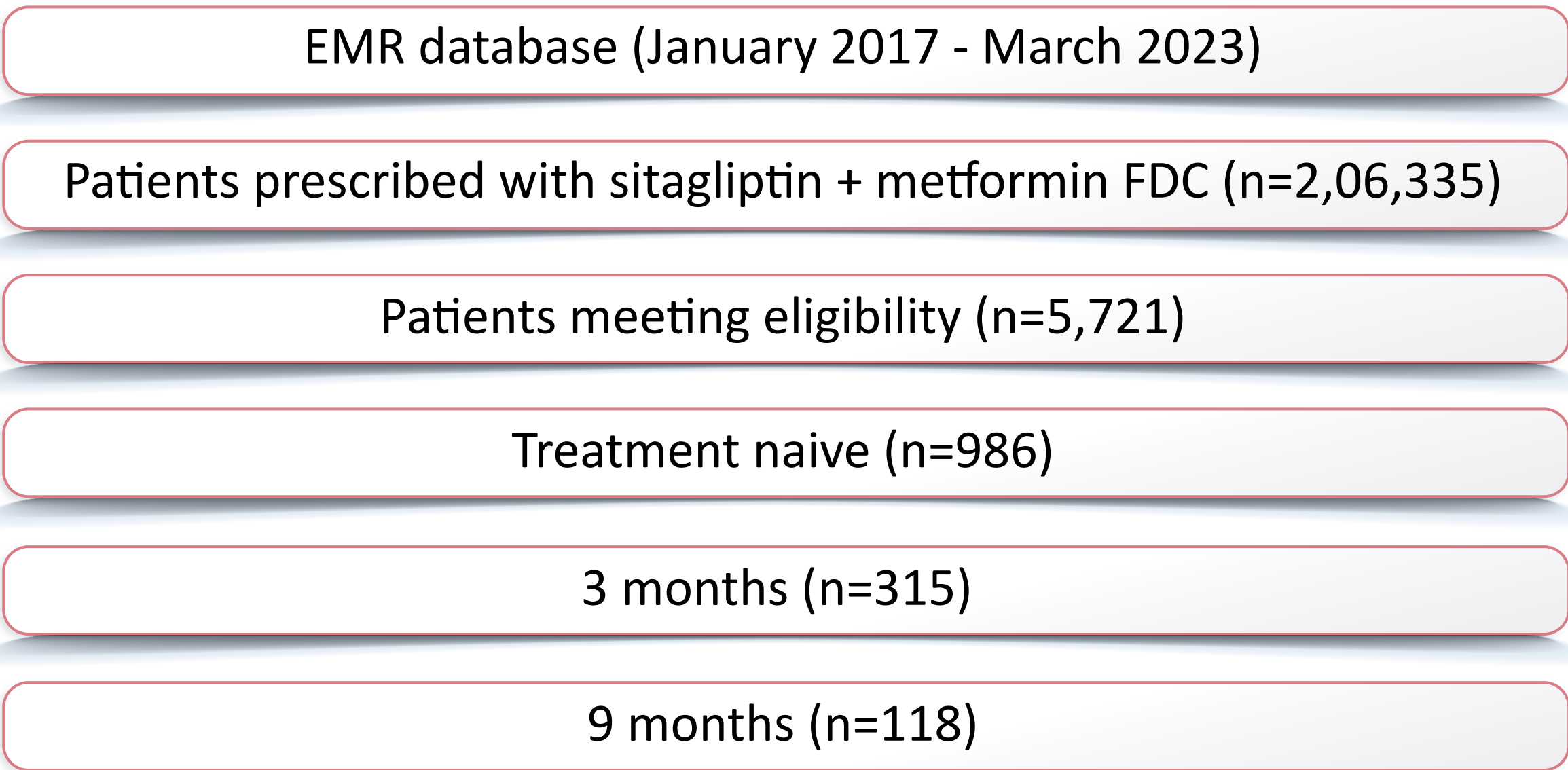


Figure 1: Study design

Efficacy Outcomes

- ▶ At baseline, the mean HbA1c was $8.94 \pm 1.78\%$ in the 3-month cohort and $8.71 \pm 1.60\%$ in the 9-month cohort
- ▶ A significant reduction in HbA1c was observed at both follow-up time points:
 - At 3 months, mean HbA1c decreased from $8.94 \pm 1.78\%$ to $7.33 \pm 1.05\%$, corresponding to a mean CFB of $-1.61 \pm 1.83\%$ ($p < 0.0001$) (Figure 2A)
 - At 9 months, mean HbA1c declined from $8.71 \pm 1.60\%$ to $7.45 \pm 1.29\%$, with a mean CFB of $-1.26 \pm 1.90\%$ ($p < 0.0001$) (Figure 2B)

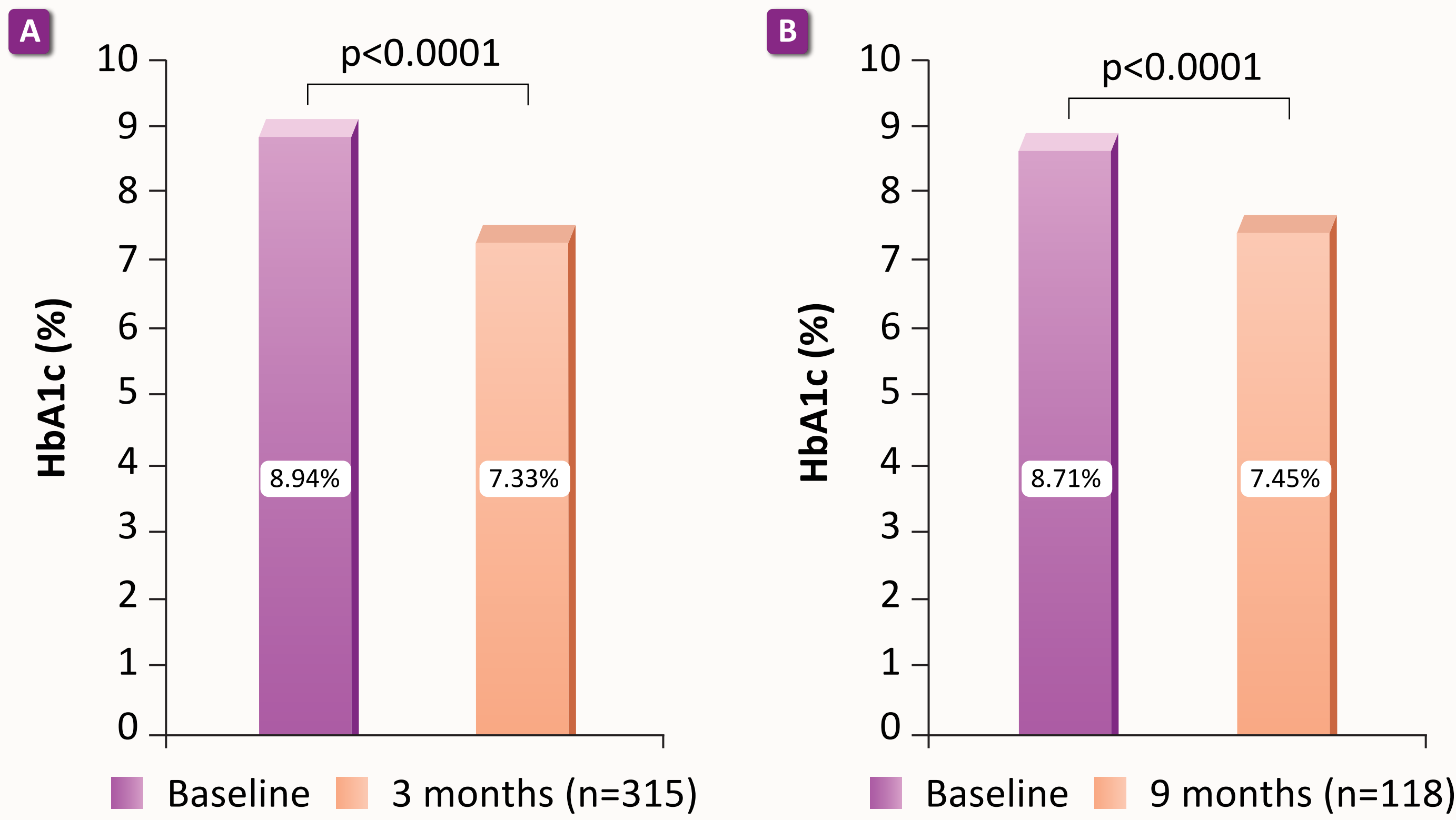


Figure 2: Mean HbA1c levels in patients with T2DM (A) at baseline and 3 months in the 3-month cohort, and (B) at baseline and 9 months in the 9-month cohort

- ▶ The proportion of patients achieving the target HbA1c level of <7% was 40.32% at 3 months, and 44.07% at 9 months (Figure 3)

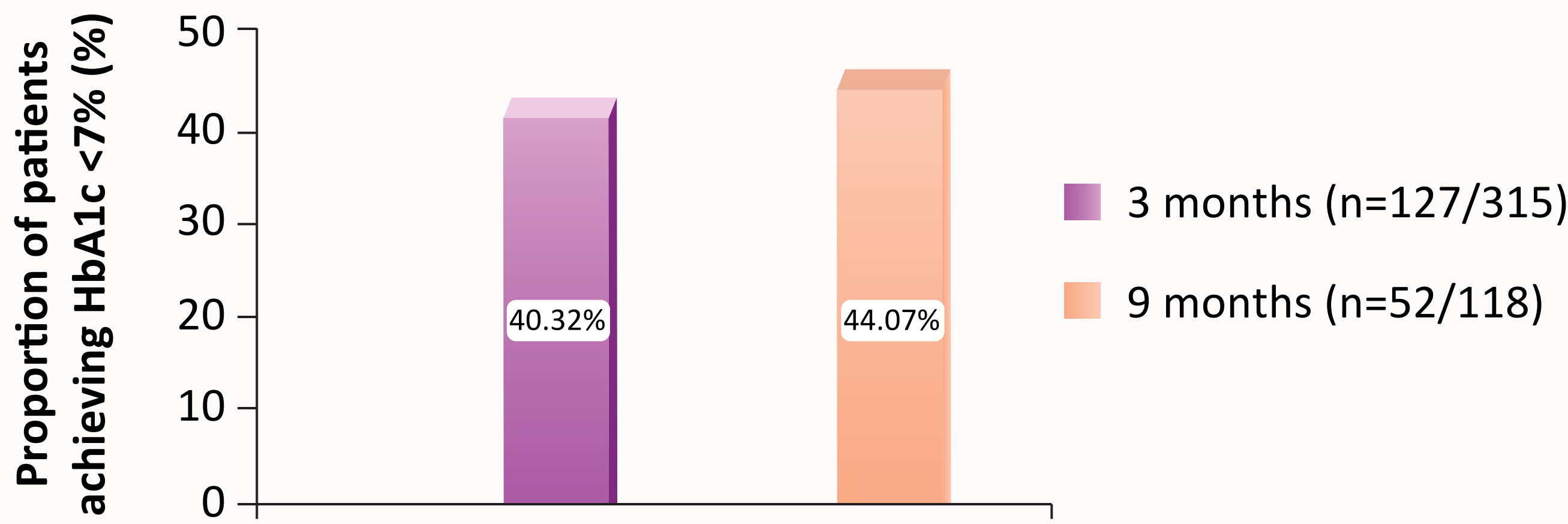


Figure 3: Proportion of patients achieving HbA1c <7% at 3 months and 9 months

LIMITATION

- ▶ This was a single arm analysis of EMR data captured on digital platform, and the database that we used does not capture safety data

CONCLUSION

- ▶ This subgroup analysis of EMR based study demonstrated effectiveness of sitagliptin and metformin FDC by significantly improving glycemic control over a 9-month period in treatment-naïve Indian patients with T2DM in a real-world setting
- ▶ These findings support the clinical utility of sitagliptin + metformin FDC as an effective and durable early treatment option for glycemic management in Indian patients with T2DM

ACKNOWLEDGEMENT

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